A Machine-Checked Thermodynamic Limit for Local Lattice Gauge Gibbs States

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Abstract.

We formalize in Lean 4 the thermodynamic limit of bounded local Gibbs expectations for a periodic lattice gauge model in a uniform Kotecky-Preiss regime. The proof treats the complete finite-volume sequence: an exact one-volume marked expansion cancels the extensive far gas algebraically, common-window terms are transported exactly, and the remaining boundary contribution is bounded by the existing volume-uniform pinned cluster tail `connectedLattice_pinned_tail_volumeUniform`. The resulting explicit Cauchy modulus tends to zero, so completeness constructs an infinite-volume positive normalized real local state. The state is invariant under each positive unit translation generator and every finite word in those generators. For d=2, SU(2), Haar probability measure, and the physical Wilson plaquette energy Re tr(U), the hypotheses are discharged throughout the explicit punctured interval 0 < |beta| <= 10^-5. We also construct a genuine centered free-boundary exhaustion and prove that its complete cofinal sequence converges to the same state as periodic boundary conditions. Finally, the normalized finite-volume two-plaquette truncated-correlation bound passes unchanged to the state under explicit eventual realization and separation hypotheses. We do not claim arbitrary boundary conditions, inverse-translation invariance, a C*-algebraic state, a continuum limit, Osterwalder-Schrader reconstruction, or progress on the continuum Yang-Mills mass-gap problem.

1. Introduction

Cluster expansions are a standard route from finite-volume polymer representations to analytic and probabilistic control in regimes of small activities. The Kotecky-Preiss criterion gives a particularly useful volume-uniform convergence mechanism for abstract polymer models [1]. For lattice gauge systems, however, a thermodynamic-limit argument must still connect the abstract cluster estimates to genuine normalized Gibbs expectations. This connection is delicate because the numerator and partition function are both extensive; neither is expected to converge separately.

The formal development studied here closes that connection for local observables. Its central design rule is to cancel the far gas exactly before applying estimates. The proof then compares two finite volumes through a common non-wrapping window. Clusters that remain inside the window agree term by term, including their Ursell coefficients and activity monomials. Clusters that leave it are support-pinned boundary tails, and are controlled uniformly in volume.

The output is not an abstract compactness statement. It is a Cauchy theorem for the complete sequence of genuine Gibbs expectations, followed by a definition of the limit using completeness. This distinction is essential: a convergent subsequence would not establish a unique thermodynamic state or boundary-condition comparison.

2. Finite-volume setting

Fix a dimension d, a measurable group G with measurable multiplication and inversion, a probability measure mu on G, a bounded measurable plaquette energy pe, and a real coupling beta. At torus extent N, a gauge configuration assigns an element of G to each positively oriented edge. The Wilson action is the sum of pe evaluated on oriented plaquette holonomies, and the Gibbs measure is the normalized tilt of the product gauge measure.

A compatible local observable records a finite support with coordinates independent of the ambient volume, a measurable bounded kernel, and a minimum admissible volume. Its realization in each sufficiently large torus reads exactly those edge coordinates. The finite Gibbs expectation E_n(O) is the genuine normalized integral of that realization at extent n+1.

> Finite local substrate.
> For every compatible local observable O, its realization is measurable and bounded; E_n is positive and normalized. Addition, multiplication, and real scalar multiplication are realized exactly at jointly admissible volumes.

3. Exact marked expansion and cancellation

Expanding each plaquette Boltzmann factor into 1 plus its Mayer activity gives a finite sum over plaquette subsets. For a local insertion O, components disjoint from the observable support factor as the ordinary polymer gas. Components touching the support form a marked activity. Mayer-Ursell inversion is then applied at one fixed volume.

```
E_n(O)
  = sum_{marked S0} markedIntegral_n(O,S0)
      * exp( clusterSum(P_n restricted to far(S0))
             - clusterSum(P_n) ).
```

This identity is exact. The extensive contribution has already cancelled between numerator and partition function. No estimate of the full partition function, and no free bulk hypothesis, appears in the thermodynamic comparison.

4. Uniform pinned boundary control

For a fixed marked set, the remaining exponent is a restriction difference. It is reindexed as a series of Ursell tuples that meet the marked region and leave the common window. Pinning an arbitrary tuple coordinate at position zero costs its cardinality n+1. The formal development records this factor explicitly and absorbs it by the unit-cardinality tilt t+epsilon+1; it is not silently discarded.

> Rooted boundary tail.
> The local correction tail is summable uniformly in the torus volume. Its proof invokes `connectedLattice_pinned_tail_volumeUniform` literally, after the exact rooted reindexing and the honest unit tilt.

The outer marked-set sum is controlled separately by decomposing a marked plaquette set into connected components. Each component is charged to the finite observable support, and a volume-uniform lattice animal bound resums the possibilities. Combining the outer tail with the normalization-correction tail yields the explicit error

```
Delta_O(q)
  = 2 * markedOuterTailBound_O(q)
    + markedSmallLayerCauchyBound_O(q,q),
Delta_O(q) -> 0 as q -> infinity.
```

5. Whole-sequence thermodynamic state

> Complete-sequence Cauchy theorem.
> Under a packaged uniform local KP regime, the sequence n |-> E_n(O) is Cauchy for every compatible local observable O. For each tolerance, the proof selects one cutoff q with Delta_O(q) below that tolerance and one explicit volume threshold above which every pair of volumes is bounded by Delta_O(q).

Completeness of the complex numbers supplies the limit. Since every finite expectation is real, the limit is real. Exact finite-volume algebra, positivity, normalization, and translation covariance pass to the limit by uniqueness of limits.

> Infinite local Gibbs state.
> The constructed functional omega_infinity is real linear, positive, and normalized on the ordered local-observable space with unit. It satisfies omega_infinity(T_i O)=omega_infinity(O) for each positive unit translation generator i and for every finite word in those generators.

The domain has an observable product, but no C*-norm or C*-completion is constructed here. Accordingly the result is described as a positive normalized local state, not as a C*-algebraic state.

6. Genuine free boundary and uniqueness at the proved scope

The periodic torus is cut along a centered seam. Plaquettes crossing that seam receive zero activity; the resulting polymer gas is proved literally equal to a restriction of the periodic gas. The observable is centered so that its support lies in the interior rather than on the cut.

Inclusion-exclusion cancels all clusters except those that meet both the local marked support and the deleted seam. Short clusters cannot do so once the common window fits. Long clusters are bounded by the same support-pinned tail. At a common volume,

```
| E_n^periodic(O) - E_n^free(centered O) |
  <= 2 * markedOuterTailBound_O(q)
     + markedSmallLayerCauchyBound_O(q,q).
```

Define the free volume index by the explicit threshold plus q. These indices are cofinal, so the periodic expectations along them converge to omega_infinity(O). The displayed difference tends to zero by squeezing. Therefore the complete explicitly indexed free sequence converges to the same value.

> Periodic versus centered free boundary.
> The full cofinal sequence of genuine centered free-box expectations converges to the same infinite value as periodic boundary conditions. This is the boundary independence formalized here; no type or theorem for arbitrary boundary conditions is asserted.

7. Truncated correlations

For local observables O and P, define the finite truncated correlation as E_n(OP)-E_n(O)E_n(P). The complete finite-volume sequence converges to the analogous expression in omega_infinity. Hence every eventual volume-uniform finite bound passes unchanged to the limit.

The formalization additionally bridges the existing normalized two-plaquette theorem to this local-observable definition. If, in all sufficiently large volumes, O and P realize a bounded measurable holonomy observable f at distinct plaquettes p_n and q_n with touching distance at least 2k, the finite normalized estimate gives

```
| omega_infinity(OP) - omega_infinity(O) omega_infinity(P) |
  <= C(d,B,beta,s,t,epsilon) * exp(-epsilon*k).
```

The eventual realization and separation data remain explicit hypotheses of this bridge. The theorem does not claim that every pair of abstract local observables is a separated two-plaquette pair.

8. Explicit non-vacuous SU(2) interval

A conditional KP theorem can be mathematically empty if its hypotheses are never instantiated, and the point beta=0 would describe only the free product measure. The development therefore constructs a concrete nonzero interval. For d=2 and SU(2), the physical plaquette energy Re tr(U) is measurable, bounded by 2, and formally proved nonconstant.

The parameters t=epsilon=eta=1/100 are fixed. On |beta|<=10^-5, the activity is bounded using the quadratic exponential remainder:

```
exp(2*|beta|) - 1 <= 2.1*|beta|.
```

Elementary rational upper bounds on the remaining exponentials then discharge the radius, smallness, unit-tilt, and marked-radius fields of the uniform regime.

> Concrete strong-coupling regime.
> For every real beta with 0<|beta|<=10^-5, the d=2 SU(2) Wilson theory has the constructed infinite local Gibbs state. Positivity, normalization, generator/finite-word invariance, and periodic-versus-centered-free convergence are therefore non-vacuous physical statements on an interval.

9. Formal verification record

Endpoint	*Role*
connectedLattice_pinned_tail_volumeUniform	*Banked volume-uniform rooted KP tail*
cauchySeq_localGibbsExpectation_kpUniform	*Complete-sequence Cauchy theorem*
infiniteLocalGibbsState	*Positive normalized local state*
su2InfiniteLocalGibbsStateOnPuncturedInterval	*Physical nonzero interval*
norm_localGibbsExpectation_sub_freeBoundary_le_kpUniform	*Same-volume periodic/free bound*
tendsto_freeBoundaryThermodynamicExpectation	*Complete free sequence has the same limit*
abs_infiniteLocalGibbsTruncatedCorrelation_le_twoPlaquette	*Finite normalized bound passes to the state*

Direct focal elaboration of all terminal modules and direct elaboration of the root `YangMillsCore.lean` terminated with exit code 0 under Lean 4.29.0-rc6. Ten headline `#print axioms` checks reported exactly `[propext, Classical.choice, Quot.sound]`.
All nine dependency HEADs matched the manifest. After supplying a process-local safe.directory setting for the ownership-mismatched checkout, the fixed-toolchain canonical root build completed with empty stderr and the literal final line `Build completed successfully (8458 jobs).` No dependency was deleted or updated and no global Git configuration was changed.

10. Scope and open directions

Proved here	*Not proved here*
Complete periodic expectation sequence is Cauchy	*Continuum or lattice-spacing limit*
Positive normalized real local state	*C*-completion or C*-state*
Positive generators and finite words	*An action of the full translation group with inverses*
Periodic and one genuine centered free exhaustion	*Arbitrary boundary conditions*
Two-plaquette bound passes under explicit geometry	*Geometry-free clustering for arbitrary local observables*
d=2 SU(2), 0<|beta|<=10^-5 instance	*Four-dimensional continuum Yang-Mills or OS reconstruction*

The next mathematical extensions are to package inverse translations as an honest local action, instantiate the eventual separated-plaquette geometry for explicit translated observables, and formulate additional physical boundary conditions as concrete polymer restrictions. These are separate theorems; none is required for the whole-sequence periodic and centered-free limits established above.

References

[1] R. Kotecky and D. Preiss, Cluster expansion for abstract polymer models, Communications in Mathematical Physics 103 (1986), 491-498. doi:10.1007/BF01211762.

[2] K. Osterwalder and E. Seiler, Gauge field theories on a lattice, Annals of Physics 110 (1978), 440-471. doi:10.1016/0003-4916(78)90039-8.

[3] L. de Moura and S. Ullrich, The Lean 4 Theorem Prover and Programming Language, CADE 28, LNCS 12699 (2021), 625-635. doi:10.1007/978-3-030-79876-5_37.

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